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Track 1: Core AI/ML & Trustworthy Systems

Scalable Random Wavelet Features

Efficient non-stationary kernel approximation with convergence guarantees

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Paper Code Author

PROBLEM

Scaling a non-stationary Gaussian process usually costs the closed-form solve.

- **Why it matters.** Speech, terrain-driven fields and step-like signals change smoothness and amplitude across the input.
- **Why it is hard.** Scalable GPs become non-stationary by learning latent functions (length-scales, warps, layers), usually variationally.
- **What is missing.** Random features keep one linear solve with guarantees, but their global sinusoids give a stationary kernel.

Our idea: put the non-stationarity in the basis, not in extra latent functions.

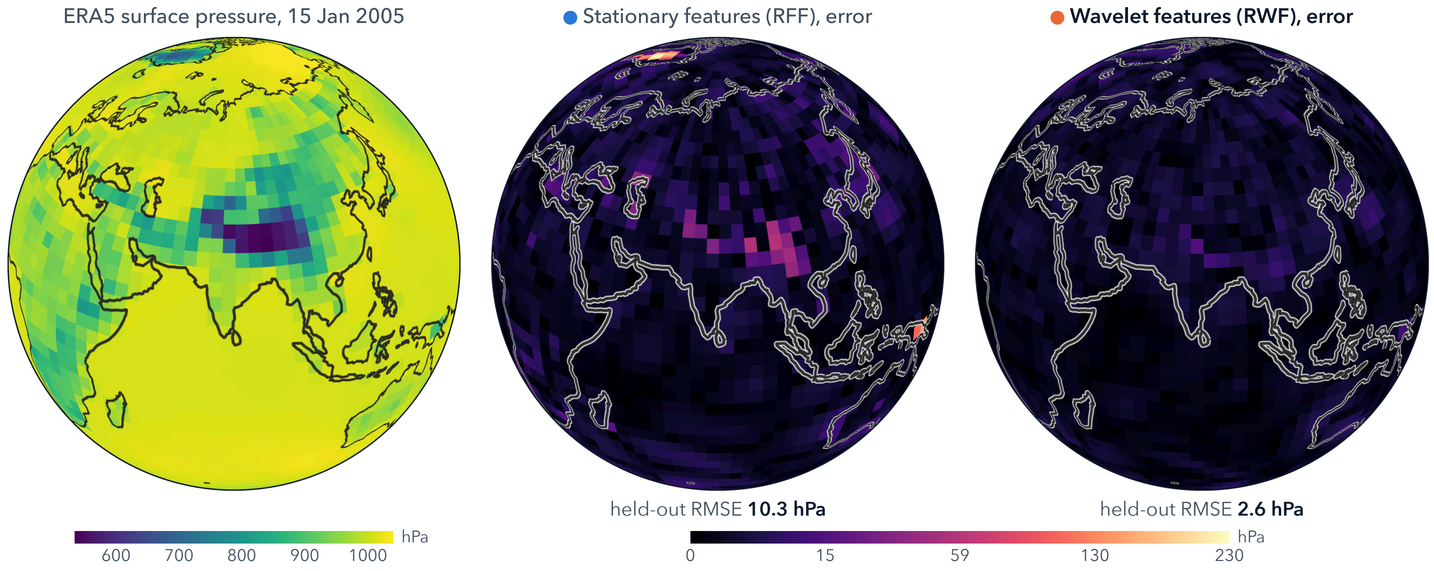
CONTRIBUTIONS

1. **Random wavelet features.** Wavelets at random scales and shifts are local: a non-stationary kernel with a closed-form solve.
2. **Guarantees.** The kernel is positive definite. Its estimator is unbiased and converges uniformly.
3. **Evidence.** Lowest error on multi-step, TIMIT, Protein and 5 of 7 UCI sets against deep and adaptive GPs; none is faster.

APPLICATION

New since the ICLR paper

Reconstructing global weather fields from a quarter of the grid



Spatial interpolation of ERA5 fields. Each field on a 48 × 96 grid is reconstructed from 1,152 simulated stations (25% of the cells) and scored on the other 3,456. Inputs are each cell's (x, y, z) position on the unit sphere, its elevation and a land mask. **A controlled comparison:** both models use 2,495 features and the same closed-form solver, so only the feature family differs. RWF uses radial Mexican-hat wavelets at four scales with a low-pass Fourier block. Errors are absolute, on a square-root colour scale.

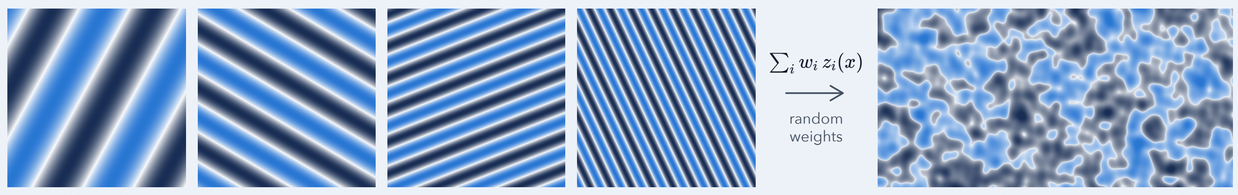
Held-out RMSE, six variables mean of 24 runs

VARIABLE	STATIONARY	RWF	CHANGE
Surface pressure, hPa	8.80	**3.02**	−66%
Temperature, K	3.09	**2.06**	−34%
Zonal wind, m/s	7.88	**5.64**	−28%
Humidity, g/kg	1.44	**1.08**	−25%
Meridional wind, m/s	7.47	**6.06**	−19%
Precipitation, log mm	1.46	**1.32**	−10%

METHOD

Random features that are local in both space and frequency

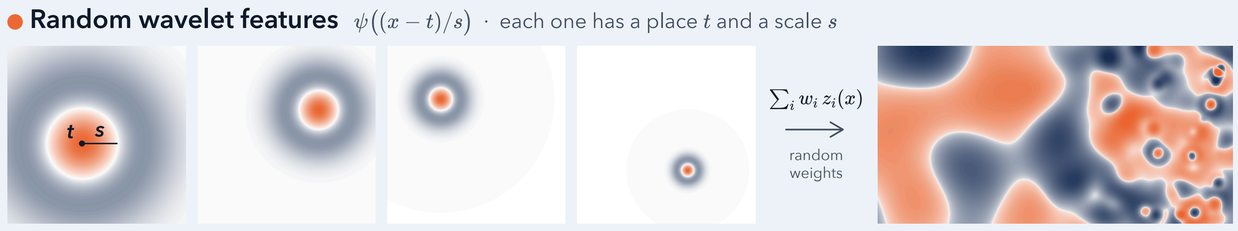
Random Fourier features $\cos(\omega^\top x + b)$ · each one spans the whole domain



Four of the D features on a 2-D input

Random sum: the same grain everywhere

Random wavelet features $\psi((x-t)/s)$ · each one has a place t and a scale s



Mexican-hat atoms at four scales and places

coarse atoms only **fine atoms too**

Random sum: smooth left, detailed right

1. **Wavelet atom.** Scale and shift a mother wavelet ψ (Mexican hat, Morlet).

$$\psi_{s,t}(x) = s^{-d/2}\,\psi\Big(\frac{x-t}{s}\Big), \qquad s>0,\;\; t\in\mathbb{R}^d$$

2. **Kernel.** Integrate atoms against a density $p(s,t)\geq 0$.

$$k(x,y) = \int_0^\infty \int_{\mathbb{R}^d} \psi_{s,t}(x)\,\psi_{s,t}(y)\;p(s,t)\;dt\,ds$$

3. **Random features.** Draw D atoms, $(s_i,t_i)\sim p$: log-uniform scales, uniform shifts.

$$z(x) = \tfrac{1}{\sqrt{D}}\big[\psi_{s_1,t_1}(x),\;\ldots,\;\psi_{s_D,t_D}(x)\big]^\top, \qquad \hat{k}(x,y) = z(x)^\top z(y)$$

4. **Inference.** Bayesian linear regression on $Z\in\mathbb{R}^{N\times D}$, solved once in closed form.

$$S_w = \big(I_D + \sigma^{-2} Z^\top Z\big)^{-1}, \qquad m_w = \sigma^{-2} S_w Z^\top y$$

THEORY

The approximation is a valid kernel and converges uniformly

THEOREM

Positive definiteness

For every non-negative density $p(s,t)$, the wavelet kernel k is positive definite on $\mathbb{R}^d\times\mathbb{R}^d$.

LEMMA

Unbiased, variance $\leq B^2/D$

$$\mathbb{E}\big[\hat{k}(x,y)\big] = k(x,y)$$
$$\mathrm{Var}\big[\hat{k}(x,y)\big] \leq B^2/D$$

THEOREM

Uniform convergence on a compact set $\mathcal{M}$

$$\Pr\Big[\sup_{x,y\in\mathcal{M}} \big|\hat{k}(x,y)-k(x,y)\big| \geq \epsilon\Big] \leq 2\Big(\frac{4\,\mathrm{diam}(\mathcal{M})\,L_z}{\epsilon}\Big)^{2d} e^{-D\epsilon^2/8B^2}$$

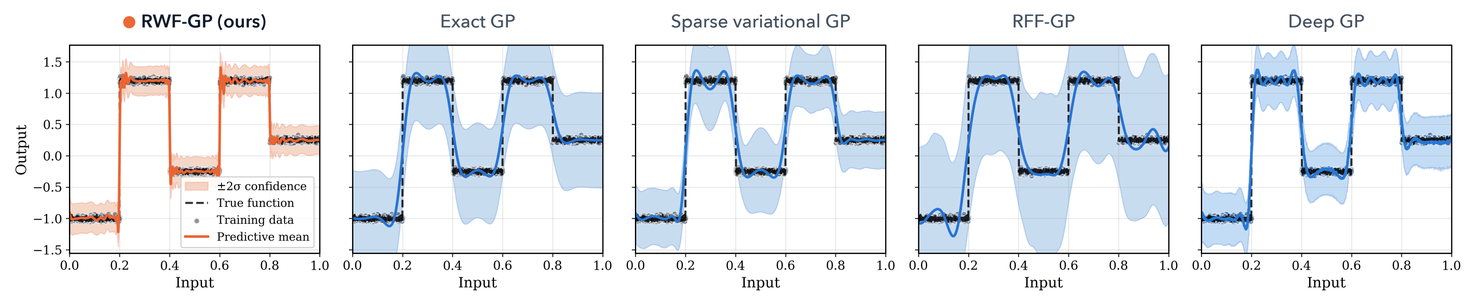
So $D = \mathcal{O}\big(\epsilon^{-2}\,d\,\log(\mathrm{diam}(\mathcal{M})/\epsilon)\big)$ features suffice.

COST

MODEL	TRAIN	PER TEST	NON-STAT.
Exact GP	$\mathcal{O}(N^3)$	$\mathcal{O}(N^2)$	by kernel
RFF-GP	$\mathcal{O}(ND^2)$	$\mathcal{O}(D^2)$	no
RWF-GP	$\mathcal{O}(ND^2)$	$\mathcal{O}(D^2)$	**yes**

RESULTS

More accurate than stationary features, cheaper than deep models

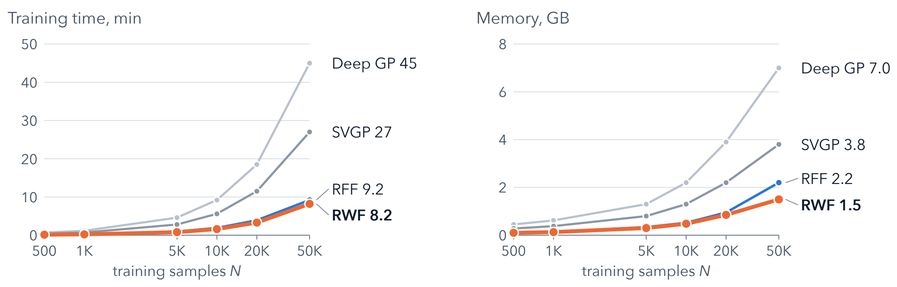


Multi-step function ($N = 4{,}200$). RWF-GP resolves every jump and keeps the uncertainty band tight. Stationary kernels over-smooth the jumps or oscillate around them.

Multi-step function mean of 5 runs, underline second best

MODEL	RMSE	CRPS	NLL	TIME
Exact GP	0.190	0.215	0.042	12
SVGP	0.231	0.392	0.123	15
RFF	0.246	0.238	0.118	11
Deep RF	0.190	0.205	−0.018	18
Deep GP	0.162	0.187	−0.268	20
Spectral mixture	0.210	0.201	0.220	17
IDD	0.107	0.143	−0.820	17
A-RKHS	0.095	0.131	−1.210	11
RWF (ours)	**0.071**	**0.112**	**−1.879**	**9**

Scaling with data multi-step function, value at 50K at line ends

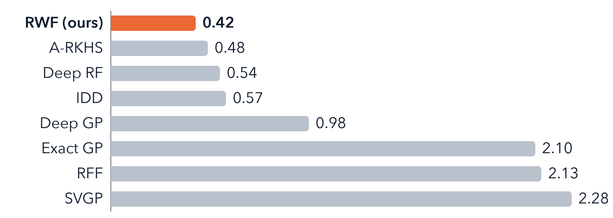


UCI regression, RMSE RWF lowest on 5 of 7

DATASET	RFF	BEST OTHER	RWF
Energy	0.66	0.48 DGP	**0.42**
Concrete	6.72	**4.20** IDD	4.45
Airfoil	5.34	3.25 A-RKHS	**3.20**
Stock	1.86	0.86 A-RKHS	**0.84**
Motion	1.60	**0.44** DRF	1.55
Kin8nm	0.41	0.09 DGP	**0.09**
Naval	0.13	0.04 DGP	**0.02**

Protein, 45K samples: RMSE 4.25 (next best 4.32).

TIMIT speech, RMSE 3,700 utterances



RWF also has the lowest CRPS, NLL and training time.

TAKEAWAYS

- Sampling wavelet scales and shifts extends random features to non-stationary kernels.
- Positive definite, unbiased, and uniformly convergent with an explicit feature count.
- Lowest error on multi-step, TIMIT, Protein and 5 of 7 UCI datasets; no baseline trains faster.
- On six ERA5 fields, swapping sinusoids for wavelets lowers RMSE by 10–66% at equal cost.

Paper Kumar & Chakraborty, ICLR 2026, arXiv:2602.00987. **Baselines** RFF: Rahimi & Recht, NeurIPS 2007 · Deep GP: Salimbeni et al., ICML 2019 · IDD: Rudner et al., ICML 2020 · A-RKHS: Shi et al., 2024 · Deep RF: Chen et al., ICLR 2025. **Data** ERA5 reanalysis, ECMWF. **Support** MoE research fellowship; ANRF grant CRG/2023/007667.